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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCBTG100 - TOOLS AND TECHNIQUES IN BIOTECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks: 35**

Remember (K), Understand (U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their Overall outcome attainability.

Part A-One Word Answer-Answer any 10 questions-Each question carries 1 marks

1. What is the process of making RNA from DNA? [K] / [CO1]
   Transcription- 1 mark
2. What are the two types of DNA ends created by restriction enzymes? [K] / [CO2]
   Blunt ends and sticky/cohesive ends- 1 mark
3. What is the starting point of DNA replication? [U] / [CO1]
   Ori (origin of replication)-1 mark
4. Which gel electrophoresis method is better for resolving small DNA fragments? [U] / [CO1]
   Poly Acrylamide Gel Electrophoresis- 1 mark
5. What are the steps involved in expressing a protein in a host cell by gene cloning? [U] / [CO2]
   Basic steps- 1 mark
6. Define bioinformatics. [K] / [CO3]
   Definition- 1 mark
7. What is recombinant DNA technology? [U] / [CO2]
   Definition- 1 mark
8. Recall the functions of qPCR. [K] /[CO1, CO4]
   Functions (Any two) - 1 mark
9. What is a cloning vector? [K] / [CO2]
   Cloning vector- definition- 1 mark
10. What is GenBank? [U] / [CO3]
    Definition-1 mark
11. Why does DNA migrate when exposed to an electric field? [U] / [CO1]
    Negative charge and electrophoretic mobility-1 mark

12. List out the steps of protein isolation. [K] / [CO4]

    Steps (minimum 3 steps-cell lysis, extraction purification)- 1 mark

Part B-Short Essay-Answer any 3 questions-Each question carries 5 marks

13. What is translation, and why is it important for protein synthesis? [U] / [CO1]

    Translation-Definition-1 mark, Importance-Transfer of genetic information coded in the DNA to protein form- explain- 4 marks

14. Comment on the role of ribosomal RNA (rRNA) in protein synthesis. [U] / [CO1]

    Role of ribosomal RNA (rRNA)- 5 marks

15. Discuss mutagenesis. [K] / [CO2]

    Mutagenesis- definition- 2 marks, explanation-3 marks

16. Compare and contrast DNA and RNA. [U] / [CO1]

    DNA- expansion- 1 mark, explanation- 1 mark, RNA- expansion- 1 mark,

    Explanation- 1 mark, Difference- (minimum 2 points) – 1 mark

17. Summarize the chromatographic techniques used in protein purification. [U] / [CO4]

    Chromatographic techniques-Any two- principle-2 marks, technique-3 marks

Part C-Essay-Answer any 1 questions-Each question carries 10 marks

18. Recall and summarize the process of PCR with reference to the components and steps in PCR [K] / [CO1, CO4]

    PCR-Expansion- 1 mark, components- 3 marks, Process-steps-6 marks.

19. Discuss the steps involved in the expression of a gene to a protein. [U] / [CO2]

    Central Dogma-DNA→RNA→Protein- 2 marks, enzymes involved in each process-2 marks, Process- steps—6 marks.